



3.3 Infrastructure

The infrastructure sector in India plays a pivotal role in the economic progress of the country, propelling its overall development. This sector encompasses a wide array of services including transportation (roads, railways, airports, and ports), energy and electricity, real estate and construction, urban infrastructure (water supply, sanitation, and waste management), and telecommunications. The Indian government has been actively promoting infrastructure growth through initiatives like the National Infrastructure Pipeline (NIP) and the Smart Cities Mission, aiming to enhance the quality of life and attract foreign investment. The sector has seen substantial investments, both from public and private sources, leading to significant improvements and modernization. However, it faces certain challenges such as regulatory hurdles, financing gaps, and the need for technological upgrades. Despite these challenges, the infrastructure sector in India is on an upward trajectory, with ongoing projects set to bolster connectivity, improve efficiency, and key enabler in helping India become a US\$26 trillion economy.⁴⁷

3.3.1 Macroeconomic trends

Key statistics

The Indian infrastructure sector is on a promising trajectory



Forecasts indicate substantial growth from approximately US\$204.06 billion in 2024 to US\$322.27 billion by 2029



This growth reflects a robust Compound Annual Growth Rate (CAGR) of 9.57% over this period⁴⁸



The Budget for 2024-25 has further reinforced this momentum with an 11.1% increase in capital investment outlay for infrastructure, totaling INR11.11 lakh crore (approximately US\$133.86 billion), which is 3.4% of India's GDP



Additionally, the budget allocation for the Railways has seen a notable 5.8% increase, reaching INR2.55 lakh crore (US\$30.72 billion) compared to the previous year⁴⁹



The infrastructure sector's contribution to the GDP is anticipated to remain significant, accounting for approximately 3.4% of India's GDP in the fiscal year 2024-25⁵⁰. This underscores the sector's pivotal role in driving the country's economic progress.

Foreign Direct Investment (FDI) continues to be a strong pillar for the sector, with total investments in construction development and infrastructure reaching approximately US\$ 60.06 billion from April 2000 to December 2023. Of this, US\$26.54 billion was directed towards construction development, while US\$33.52 billion was invested in infrastructure.

Looking ahead, India is committed to improving its Logistics Performance Index ranking to 25 and reducing logistics costs from 14% to 8% of GDP over the next five years. These ambitious goals highlight the government's strategic focus on enhancing efficiency and competitiveness in the sector.



B Employment scenario

The infrastructure sector is a vital contributor to India's employment landscape with an estimation of 9.8 million jobs to be created in FY25⁵¹. This underscores the sector's critical role in providing job opportunities across the country.

In the construction sector, the percentage of workers has increased to 13.0% in 2022-23 as compared to 12.1% in 2020-21⁵². This scenario opens significant potential for new entrants into the workforce, particularly as the sector continues to expand.

The Union Budget for 2024-25 has reinforced this commitment by announcing plans to upgrade 1,000 ITIs using a hub-and-spoke model⁵³. This initiative is designed to align course content with current industry skill requirements and introduce new, relevant courses, thereby enhancing the quality of technical education and ensuring that graduates are well-prepared for careers in the infrastructure sector.

A new centrally sponsored scheme under the Prime Minister's package, set to be introduced in collaboration with state governments and industry partners, aims to skill 20 lakh first timers over the next five years. This ambitious initiative underscores the government's strong commitment to bridging the skill gap and ensuring a steady supply of skilled labor for the infrastructure sector.

While there is a recognized need for skilled labor, particularly certified professionals and those with specialized skills in operating advanced construction equipment and technologies, this also creates a promising landscape for upskilling and training initiatives. Addressing this skilled labor requirement is poised to unlock even greater potential in the sector, contributing to its sustained growth and the overall economic development of the country.

C Key drivers of growth and government schemes

Government of India has made a substantial commitment to infrastructure development, significantly increasing the budget allocation to INR5 lakh crore (approximately US\$60 billion) for FY2024, up from INR3.7 lakh crore in FY2023. This investment aligns with the objectives of the National Infrastructure Pipeline (NIP), which oversees projects worth INR108 trillion (approximately US\$1.3 trillion) at various stages of implementation, all aimed at enhancing the country's infrastructure.

The sector is also embracing emerging technologies such as smart construction, digital infrastructure, and renewable energy solutions, which are set to drive both efficiency and sustainability in infrastructure development.

Several major government initiatives are in place to support sustainable infrastructure development:

- ▶ **National Infrastructure Pipeline (NIP):** Focused on improving infrastructure quality while promoting green practices, the NIP plays a crucial role in the nation's long-term growth strategy.
- ▶ **PM Gati Shakti:** This initiative is accelerating the construction of national highways, with a target to complete 200,000 km by 2025, significantly enhancing connectivity.
- ▶ **UDAN Scheme:** Designed to improve air connectivity to underserved regions, the UDAN Scheme has operationalized 415 routes and 66 airports, fostering regional development.
- ▶ **Sagarmala Project:** Aimed at enhancing port connectivity and logistics, the Sagarmala Project has attracted significant investments, boosting cargo handling capacity and overall maritime infrastructure.

- **India Infrastructure Project Development Fund (IIPDF):** Supporting Public-Private Partnership (PPP) projects, the IIPDF provides financial assistance for project development expenses, encouraging private sector participation in infrastructure projects.
- The government is proactively addressing these areas to ensure sustained progress. The establishment of the Urban Infrastructure Development Fund (UIDF) with an annual outlay of INR10,000 crores is a testament to these efforts, targeting the improvement of urban infrastructure and quality of life in cities.
- To further bolster infrastructure development, the government has introduced diverse financial instruments aimed at mobilizing substantial resources for various projects. These include Infrastructure Debt Funds (IDFs), Real Estate Investment Trusts (REITs), and tax-free bonds, all contributing to a more dynamic and resilient financing ecosystem.

Collectively, these comprehensive policies and initiatives underscore the government's strong commitment to advancing India's infrastructure landscape, overcoming existing challenges, and paving the way for sustained economic prosperity.

3.3.2 International benchmark

India has achieved remarkable progress in its infrastructure sector, as evidenced by its improved ranking in the World Bank's Logistics Performance Index (LPI). In 2023, India advanced to 38th place out of 139 countries, up from 44th in 2018. This significant improvement reflects the effectiveness of key government initiatives such as PM Gati Shakti and the National Logistics Policy, both of which are focused on enhancing the nation's soft and hard infrastructure. Furthermore, India has made substantial progress in the International Shipments category, climbing from 44th in 2014 to 22nd in 2023, driven by successful modernization and digitalization efforts within the logistics sector⁵⁴.

In addressing job roles and skills within the infrastructure sector, international best practices offer valuable insights:



Long-term strategic planning: Countries like the UK and Australia emphasize the importance of long-term strategic infrastructure planning, which aligns workforce skills with future needs, ensuring the sector remains adaptable and prepared for upcoming challenges⁵⁵.



Skill development roadmaps: The UK's National Infrastructure Commission provides a comprehensive roadmap for future skill requirements, allowing educational institutions to tailor their curricula to meet these evolving demands.



AI and efficiency: Germany and the USA are at the forefront of leveraging artificial intelligence (AI) to optimize construction processes and project management, enhancing efficiency and reducing costs.



Sustainable practices: The Netherlands and Scandinavian countries are leaders in integrating green building technologies and renewable energy into infrastructure projects. This focus on sustainability is driving the development of green skills through specialized training programs, creating new job roles such as sustainability consultants and green building specialists.

India's continued emphasis on adopting such best practices, coupled with its commitment to modernization and sustainability, positions the nation to further enhance its infrastructure sector while simultaneously fostering the development of a skilled, future-ready workforce.

3.3.3 Sectoral trends

Emerging technologies in the sector

The infrastructure sector is undergoing a dynamic transformation, driven by the adoption of Industry 4.0 technologies. These key emerging technologies are setting the stage for enhanced efficiency, safety, and sustainability across various infrastructure projects:



Internet of Things (IoT): The IoT facilitates a network of interconnected devices and sensors that collect and exchange data, enabling real-time monitoring and decision-making in infrastructure projects. For example, real-time monitoring allows for the early identification of potential issues, enabling proactive maintenance and minimizing downtime.



Big data analytics: By analyzing large sets of data, big data analytics enhances forecasting, risk management, and operational efficiency, leading to more

informed decisions in infrastructure development.



Autonomous robots and drones: These technologies are increasingly being used for surveying, inspection, and maintenance tasks, improving safety and efficiency on construction sites.

In ports, digitalization is revolutionizing operations and boosting efficiency, while in the power sector, AI and machine learning are optimizing energy generation and grid security. The transportation sector is benefiting from AI systems that enhance road safety and monitor infrastructure conditions, and water management is seeing improvements through real-time monitoring and predictive analytics. In the oil and gas industry, digital communication technologies are enhancing asset management and predictive maintenance.

While the integration of these advanced technologies presents challenges, such as the need for a skilled workforce and the implementation of robust cybersecurity measures, these are also opportunities for growth and innovation. As India continues to invest in its infrastructure, fostering public-private partnerships and enhancing digital connectivity will be pivotal in driving these advancements forward, ultimately improving the quality of life for its citizens.

B Workforce trends

According to the National Skill Development Council (NSDC), the real estate sector is a major employer, which includes both skilled and unskilled labor. The

infrastructure sector complements this by employing a large section of the workforce. Among the 71 million individuals working in the construction industry, there are approximately 4.4 million core skilled workers, including engineers, technicians, and clerical staff, alongside 6.9 million employees who have received vocational training.

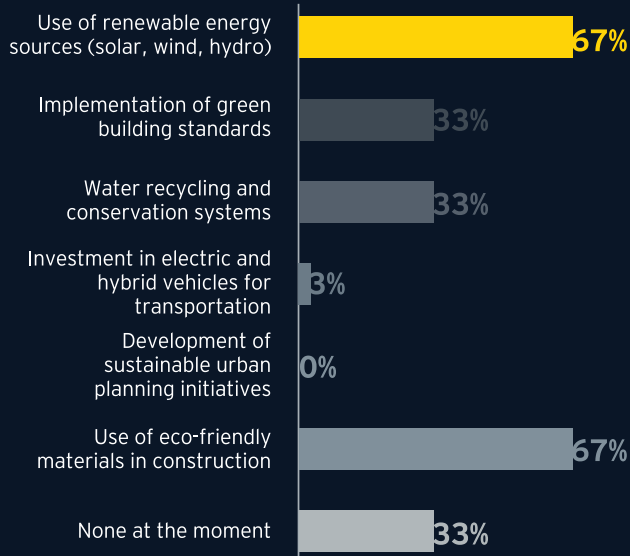
The demand for freshers across various sectors has increased by 5% over the past six months, signaling a positive recovery in hiring trends. This uptrend is bolstered by improvements in the IT sector, which also has a beneficial impact on infrastructure-related roles. As the industry evolves, specialized certifications such as Project Management Professional (PMP) and LEED accreditation are becoming increasingly valuable for roles focused on sustainability and project delivery.

While there are challenges in workforce development, particularly in bridging the skill gap between current capabilities and the needs of modern infrastructure projects, these challenges also present opportunities for growth. The gap is especially pronounced in areas related to sustainable practices and advanced technologies, where there is significant potential for upskilling. The National Federation of Independent Business (NFIB) has noted that 43% of business owners report job openings that are difficult to fill, underscoring the need for targeted training and development programs.

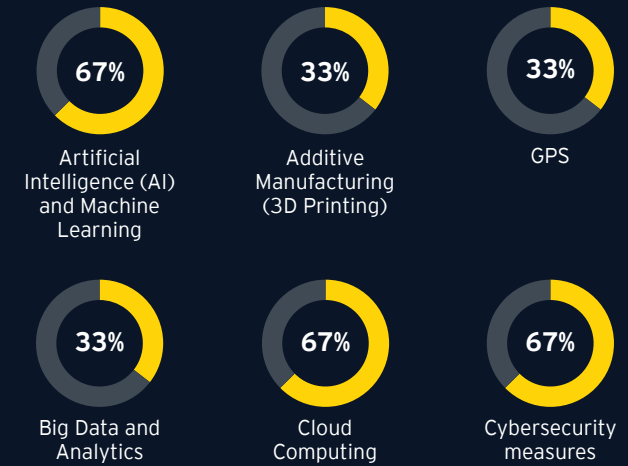
Retention remains an area for improvement, as high turnover rates can disrupt project continuity and increase training costs. However, addressing these challenges with strategic workforce development initiatives will enhance the sector's resilience and contribute to its ongoing success.



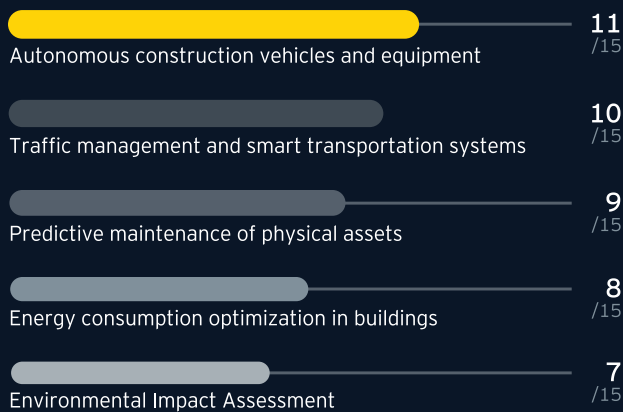
Sustainable practices the organizations in the sector have adopted



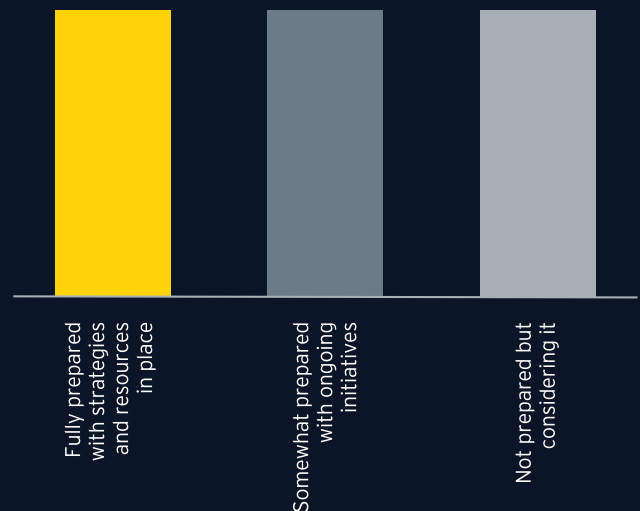
Utilization of any of the Industry 4.0 technologies



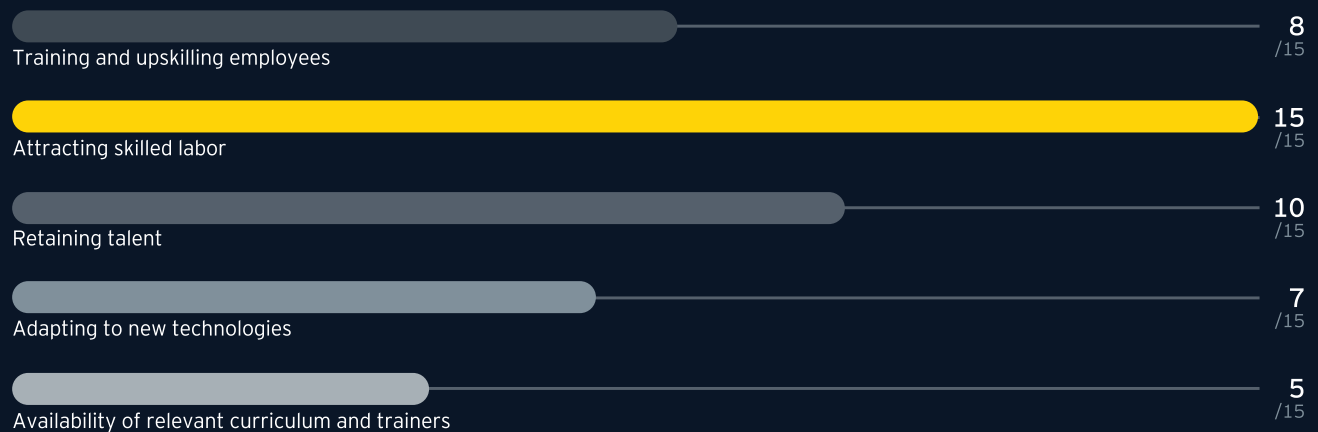
Areas of infrastructure sector where AI is expected to have the greatest impact



Preparedness level of organizations to adopt AI and Gen AI technologies



Biggest challenge in workforce development in infrastructure operations



3.3.4

Future of jobs: insights from industry leaders

Emerging job roles in the sector

Based on the survey conducted with industry leaders in the infrastructure sector, several emerging job roles are anticipated within the next decade. These roles are primarily driven by advancements in technology, sustainability and the increasing complexity of infrastructure projects. The following key job roles are expected to see significant growth:

- ▶ **AI and Machine Learning specialists:** With the increasing adoption of AI and machine learning, there will be a high demand for specialists who can develop, implement, and manage these technologies within the infrastructure sector as per 67% of the respondents.
- ▶ **Renewable energy technicians:** 67% of the respondents believe that as the focus on sustainability intensifies, there will be a growing need for technicians skilled in the installation, maintenance and optimization of renewable energy systems.
- ▶ **Sustainable infrastructure engineers:** Engineers with expertise in sustainable design and green technologies will be crucial as the sector moves towards more environment friendly practices as per 33% of the respondents.
- ▶ **Smart City planners:** 33% of the respondents believe that the development of smart cities will create a demand for planners who can integrate IoT, AI, and other advanced technologies into urban infrastructure.

Impact of GenAI, AI, sector-specific technology, international mobility, DPI, climate, and green jobs on the job and skilling ecosystem

The infrastructure sector is on the cusp of significant transformation due to several technological and environmental factors:

- ▶ **AI and GenAI:** AI is expected to have a moderate impact on the infrastructure sector over the next two years, with key areas including autonomous construction vehicles, traffic management systems, and predictive maintenance. However, challenges such as the lack of technical skills (100% of respondents' observation) and high costs of implementation (33% of respondents' observation) may slow adoption.
- ▶ **Sector-specific technology:** Technologies like cloud computing and cybersecurity are already in use by

67% of respondents, and there is a growing interest in integrating more advanced tools like AI, 3D printing, and smart sensors. The sector is still lagging in the adoption of IoT and robotics, which presents opportunities for future growth.

- ▶ **Climate and green jobs:** The sector's move towards sustainability is evident, with 67% of respondents using renewable energy sources and eco-friendly materials in construction. However, the creation and filling of green jobs face challenges such as the lack of qualified candidates and resistance to changing traditional practices, as 1/3rd of the respondents observing this.
- ▶ **International mobility:** As infrastructure projects become more global, skills related to international trade and logistics are becoming increasingly important, with 67% of respondents recognizing the need for these competencies.

Suggested programmatic and policy interventions

To meet the evolving needs of the infrastructure sector, both industry leaders and the government need to take the following actions:



Industry initiatives

- ▶ **Reskilling and upskilling:** With the anticipated impact of AI and other technologies, 67% of the respondents agree reskilling and upskilling programs to be essential to prepare the workforce for new roles and responsibilities
- ▶ **Sustainability practices:** 67% of the respondents confirmed that their organization has adopted sustainable practices such as use of renewable energy sources (solar, wind, hydro) and eco-friendly materials in construction followed by water recycling and implementation of green building standards

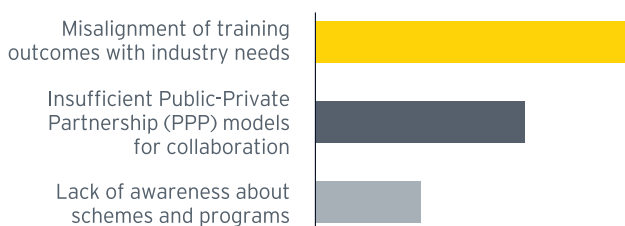


Government Intervention

- ▶ **Financial incentives:** 100% of respondents identified the need for financial incentives for green investments. The government should introduce subsidies and tax credits to encourage the adoption of sustainable practices

- Support for technology adoption: While 67% of respondents find current policies somewhat effective, there is a clear demand for more targeted support, especially in providing grants for new technology and training employees
- 100% of the respondents believed that the skilling institutions should focus on sector specific technology interventions

Challenges infrastructure companies face in collaborating with educational and skilling institutions



The future of jobs in the infrastructure sector is heavily influenced by advancements in AI, sustainability and global mobility. There is a clear need for strategic interventions from both the industry and the government to address skill gaps, promote technology adoption, and ensure sustainable development. The anticipated job roles and required skills highlight the evolving landscape of the sector, where technology and sustainability will play pivotal roles in shaping the workforce of tomorrow.

3.3.5

Overall impact on job roles and skills in the sector

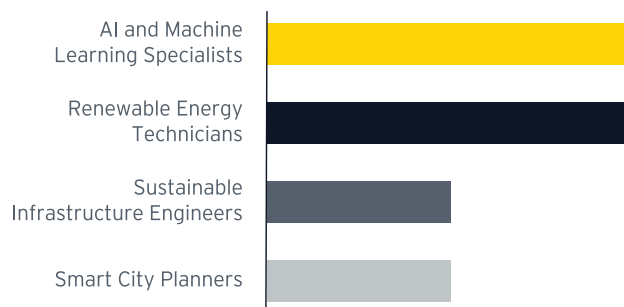
As the infrastructure sector continues to integrate new technologies and sustainable practices, job roles are undergoing significant evolution. There is a growing demand for positions centered on project management, data analysis, and environmental compliance, reflecting the sector's shift towards modernization and sustainability.

New roles, such as data centre engineers and site reliability engineers (SREs), are emerging, requiring specialized expertise in cloud technologies and automation. While the rise of automation and artificial intelligence (AI) may lead to a reduction in jobs involving routine and repetitive tasks, such as manual server provisioning and configuration management, it also

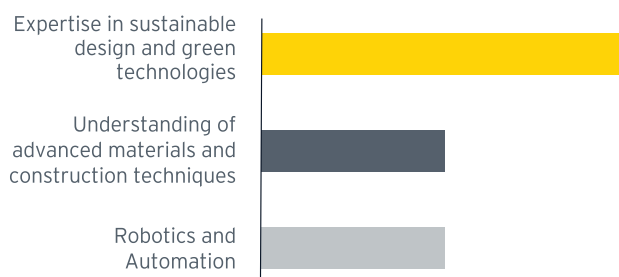
opens the door to new opportunities. The demand for skills in AI, automation, and sustainable practices will increase, highlighting the need for continuous learning and adaptation within the workforce.

As automation streamlines manual tasks, the emphasis will shift towards skills that are uniquely human and difficult for machines to replicate, such as problem-solving, critical thinking, and adaptability. Despite the challenges posed by technological advancements, the ongoing investment in infrastructure and the transition to sustainable practices are expected to generate long-term employment opportunities, particularly for those with the right skills and qualifications.

Emerging job roles in infrastructure sector



Skill sets that will be most in demand for future infrastructure job roles



This evolving landscape underscores the critical importance of preparing the workforce to meet the future demands of the infrastructure sector, ensuring that they are well-equipped to thrive in a rapidly changing environment.